

ANALOG & DIGITAL CIRCUITS

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Computer Science Department, TUCN

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Assessment

- **Lab** test + activity during labs (uploading your solutions, participating in discussions, explanations, etc)
- Written **Exam** (theory, problems)
40% laboratory + 60% final exam
Grading constraints: minimum of 5 for lab and exam/exams
- Module Credits: 4

Bibliography

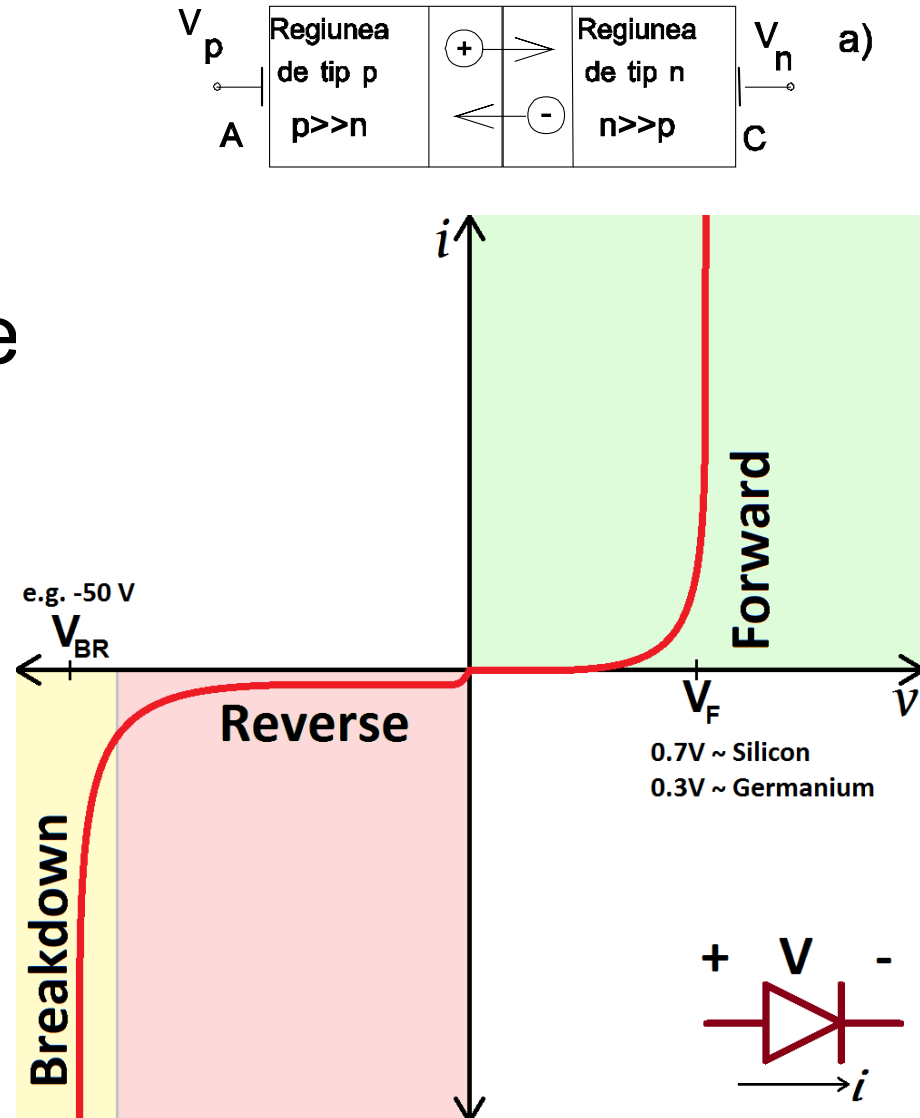
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- V. Dădârlat, “Circuite și dispozitive numerice”, Ed. Mediamira, Cluj-Napoca, 1999
- A. Peculea, B. Iancu, V. Dadarlat, S. Buzura, "Analog and digital Circuits. Practical applications", U.T. Press Cluj-Napoca, 2020
- A. Sedra, K.C. Smith, T.C. Carusone, Vincent Gaudet, “Microelectronic Circuits”, Oxford University Press, 8th edition, 2019.
- P. Horowitz, W. Hill, “The Art of Electronics 3rd Edition”, Cambridge University Press, 2015
- W. Oskay, E. Schlaepfer, “Open Circuits: The Inner Beauty of Electronic Components”, No Starch Press, 2022
- **Moodle page – lectures**

L1 Content

- Electric signal - basics
 - Electric pulse: definition, parameters, pulse generating using elementary signals
- RC circuits (pulse transformers)
 - RC circuits response to elementary signals

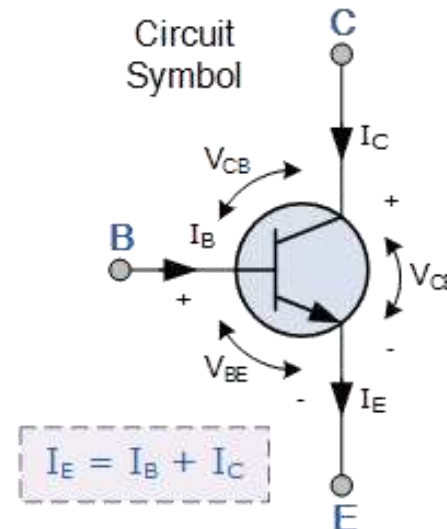
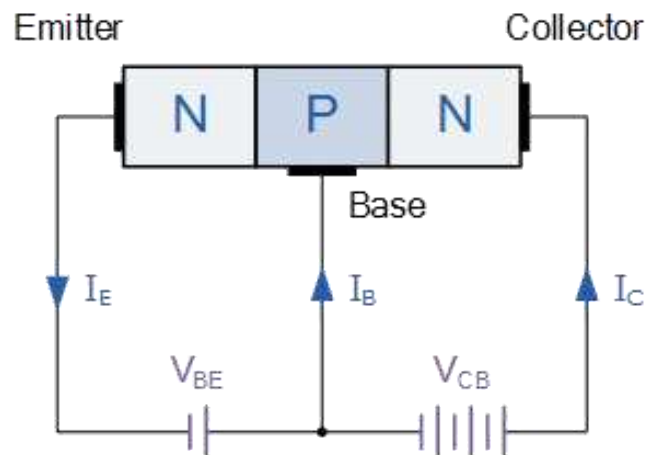
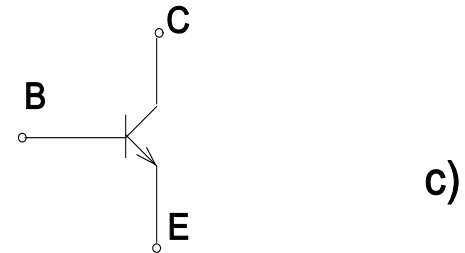
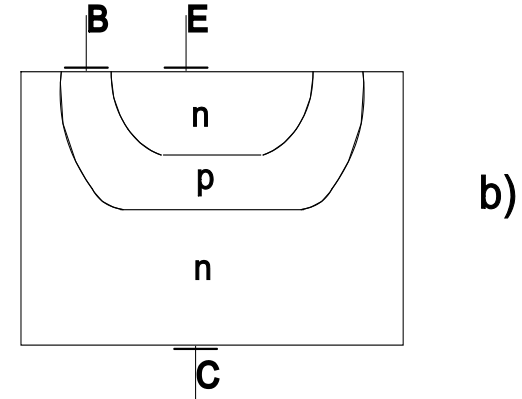
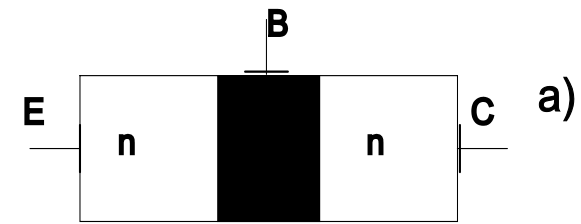
L2 Content

- The switching regime of the semiconductor device
 - Diode
 - Bipolar Transistor
 - Unipolar MOS Transistor



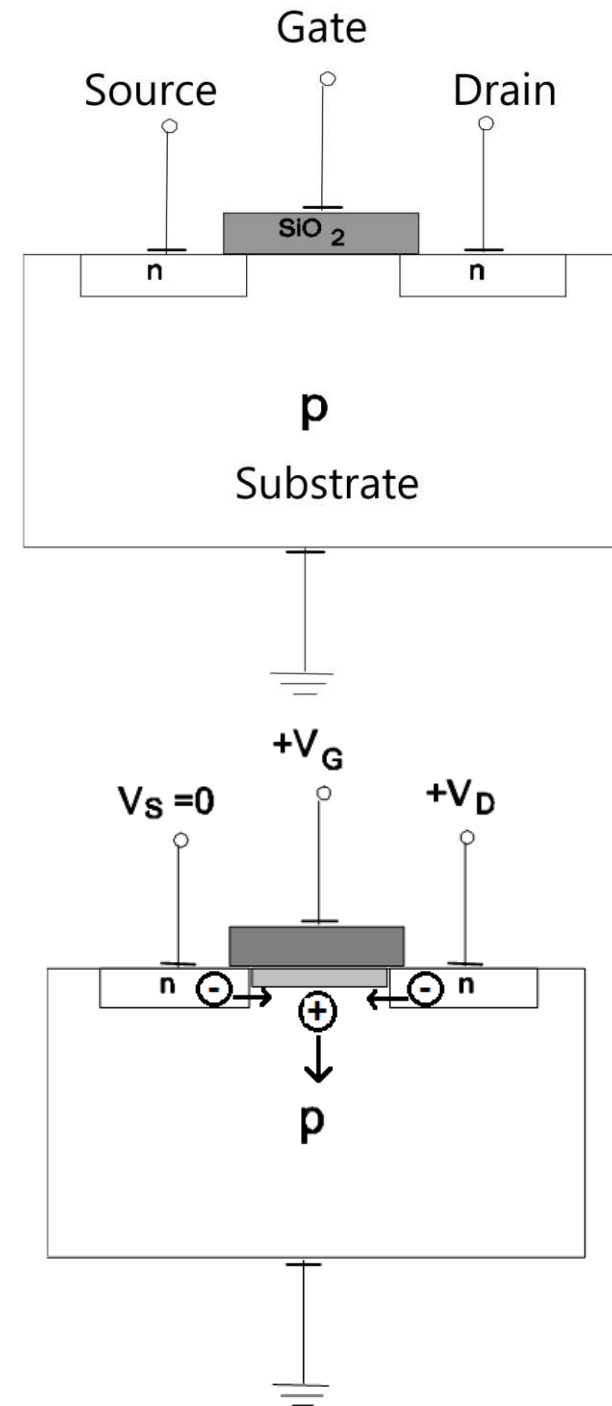
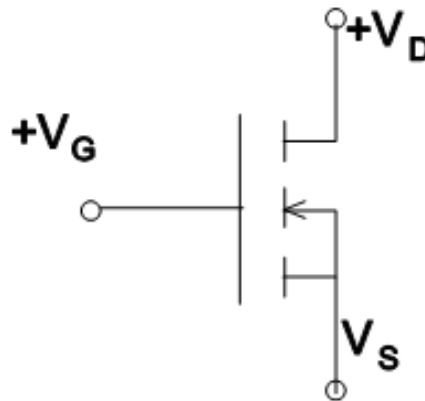
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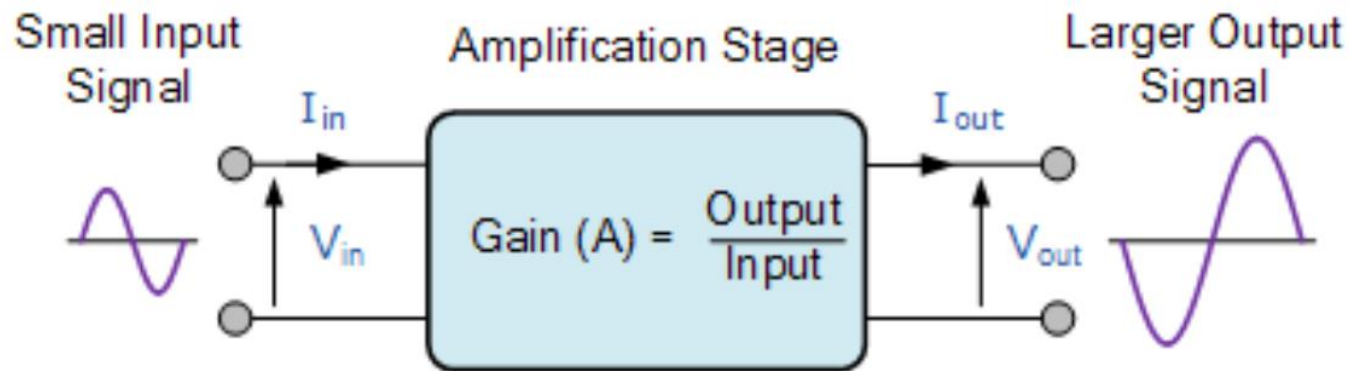
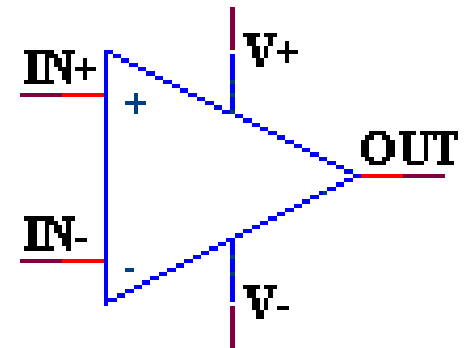
L2 Content

- The switching regime of the semiconductor devices
 - Diode
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L3 Content

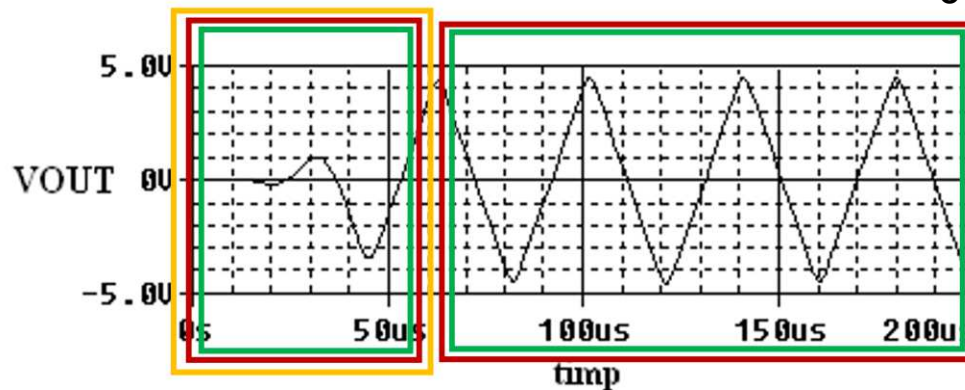
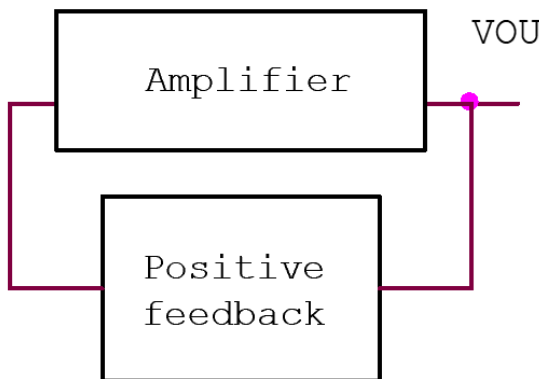
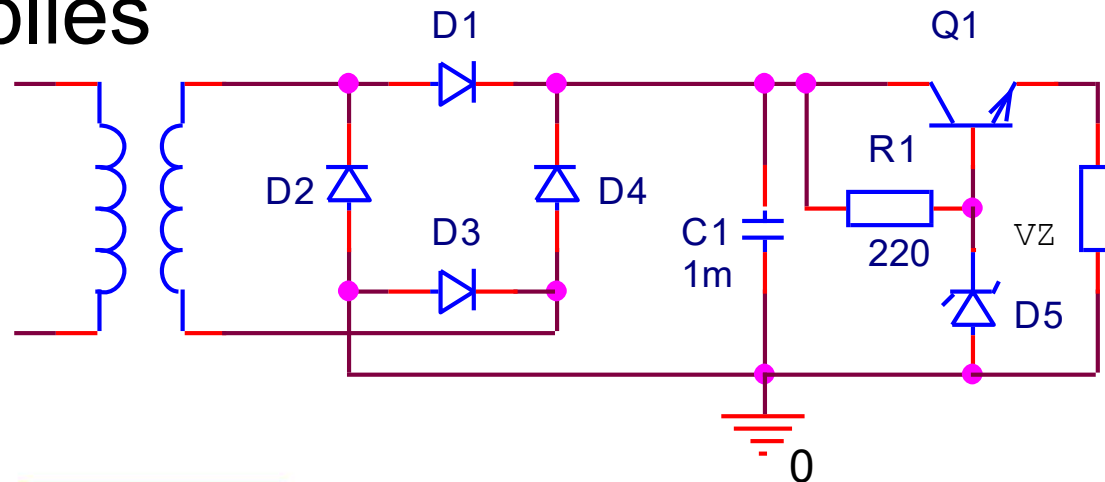
- Operational Amplifiers (Opamps)
- DC Power Supplies
- Oscillators



L3 Content

- Operational Amplifiers (Opamps)
- DC Power Supplies
- Oscillators

Voltage regulator



L4 Content

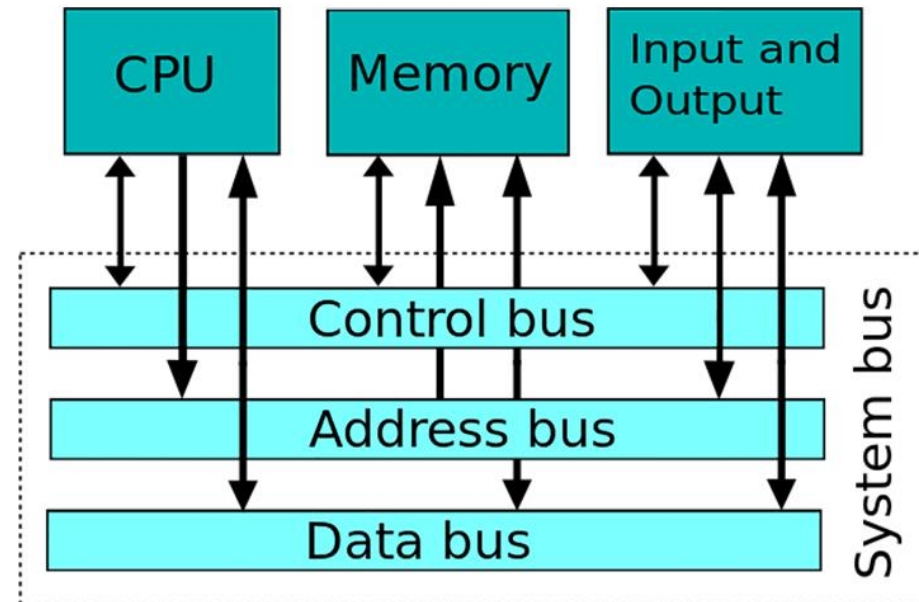
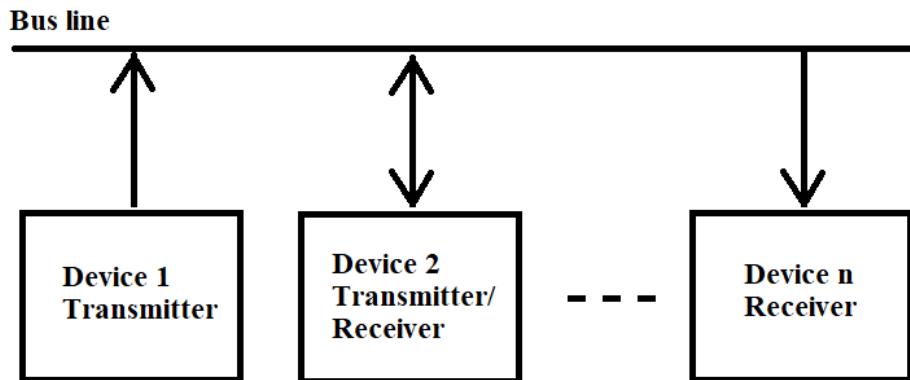
- Digital Circuits
- Parameters of integrated logic circuits
 - Static transfer characteristic
 - Noise margins
 - Fan-out & fan-in
 - Propagation time
 - Power Dissipation
- TTL logic circuits

L5 Content

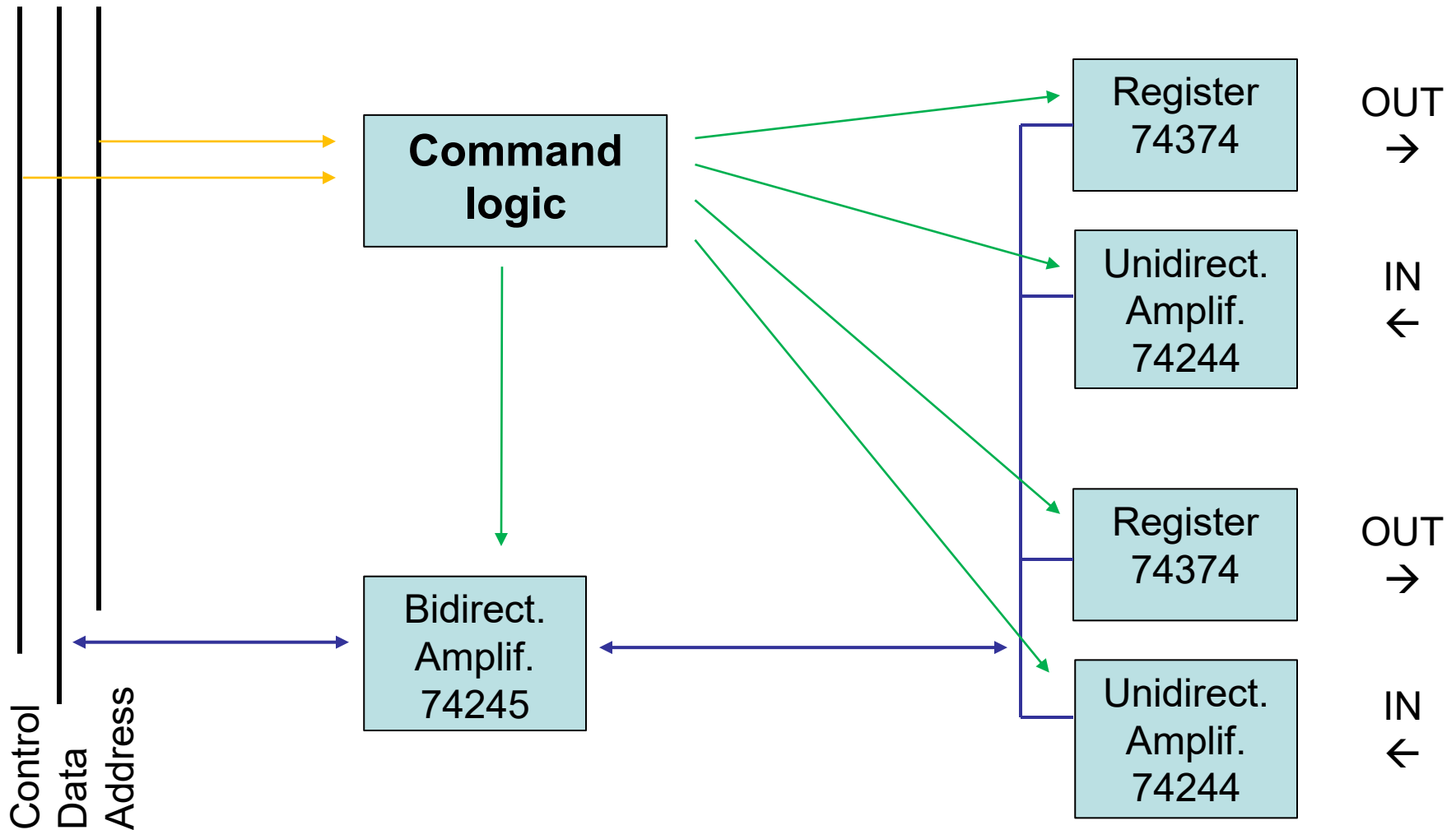
- NMOS Digital Circuits
 - Introduction
 - Static NMOS circuits
- CMOS Logic Integrated Circuits
 - Introduction
 - CMOS Inverter
 - Parameters of CMOS circuits
 - Circuits for protection
 - Output stage for CMOS circuits
 - Buffering circuits

L6 Content

- Building a Bus Using Logic Circuits
 - Introduction
 - Open Collector Logic Circuits
 - Three State Logic

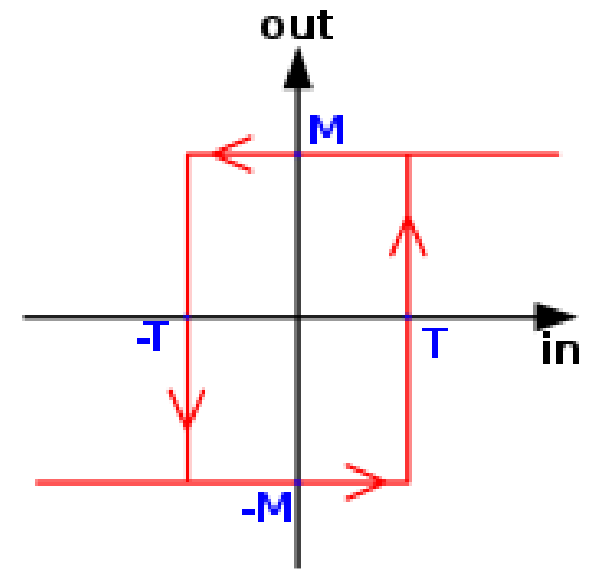
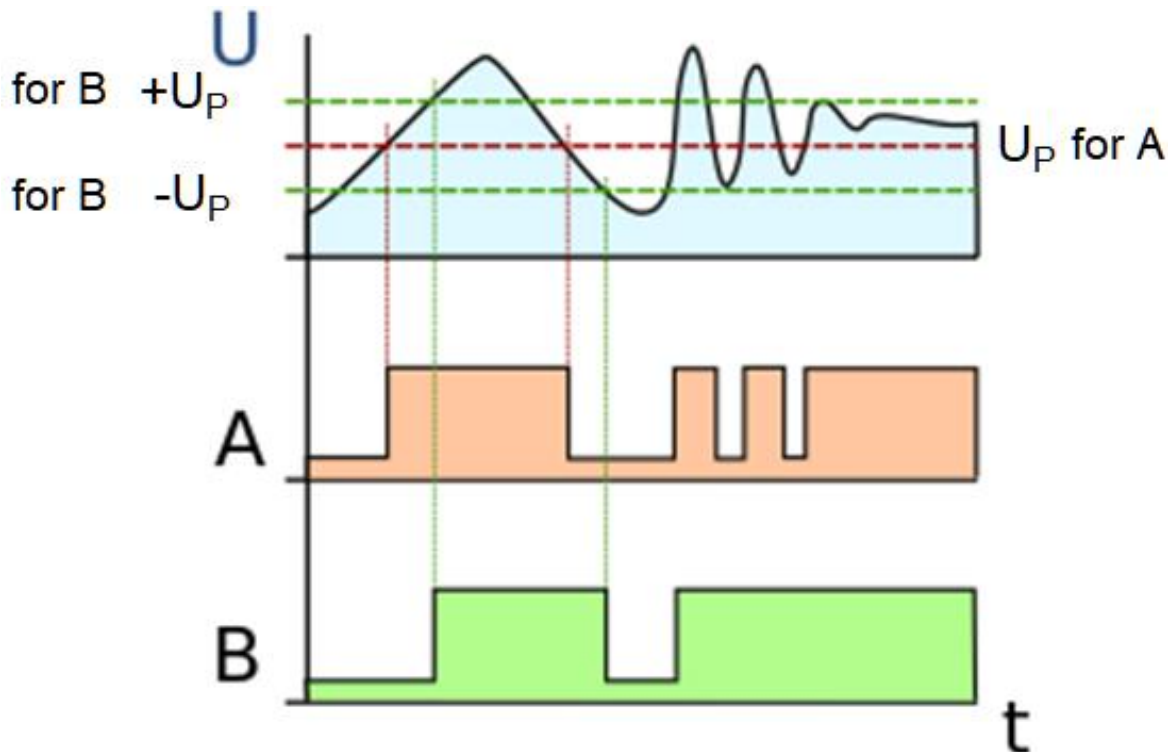
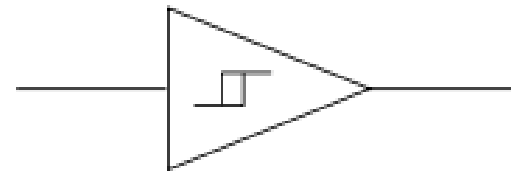


- 2 Input ports + 2 output ports
- 8 address lines (A7 ... A0)
- 8 data lines (D7 ... D0)
- 2 command signals (RD\, WR\)



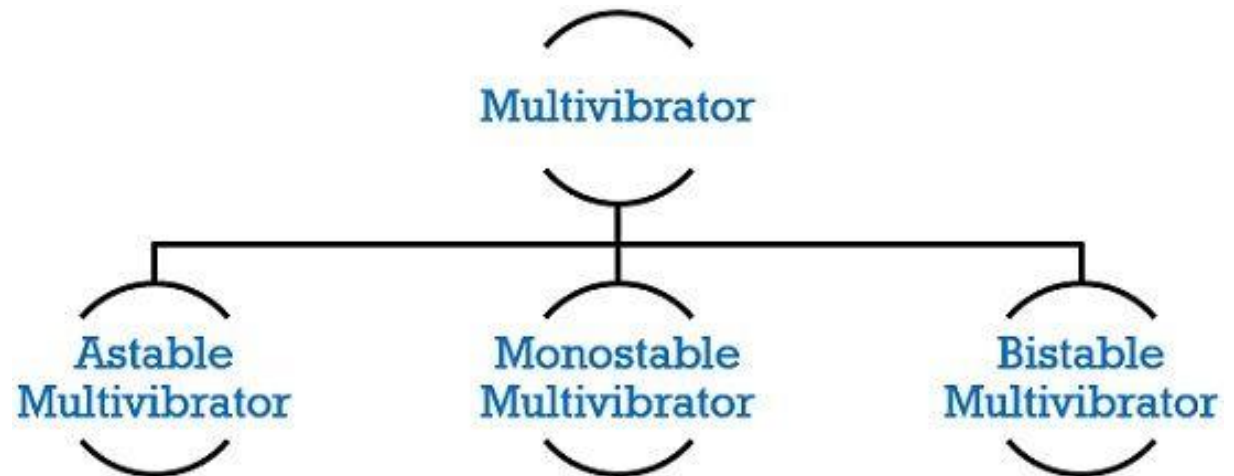
L7 Content

- Schmitt Trigger (ST) Circuits



L8 Content

- Positive feedback circuits: multivibrator circuits
 - Bistable
 - Monostables
 - Astables



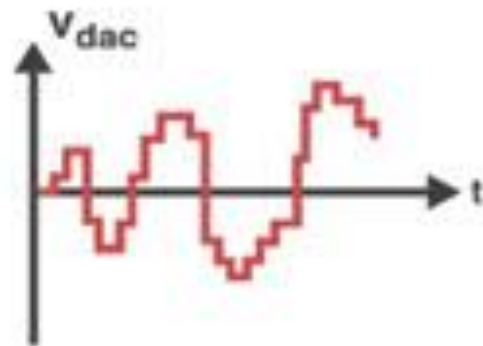
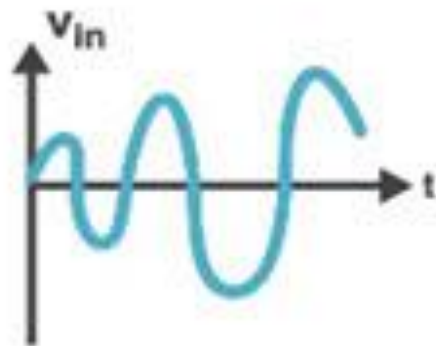
L9 Content

- Semiconductor memories
 - Volatile memories (RAM, CAM)
 - static (SRAM Static Random Access Memory)
 - dynamic (DRAM Dynamic Random Access Memory)
 - Non-volatile memories (ROM)
 - PROM
 - REEPROM
 - Flash

L10 Content

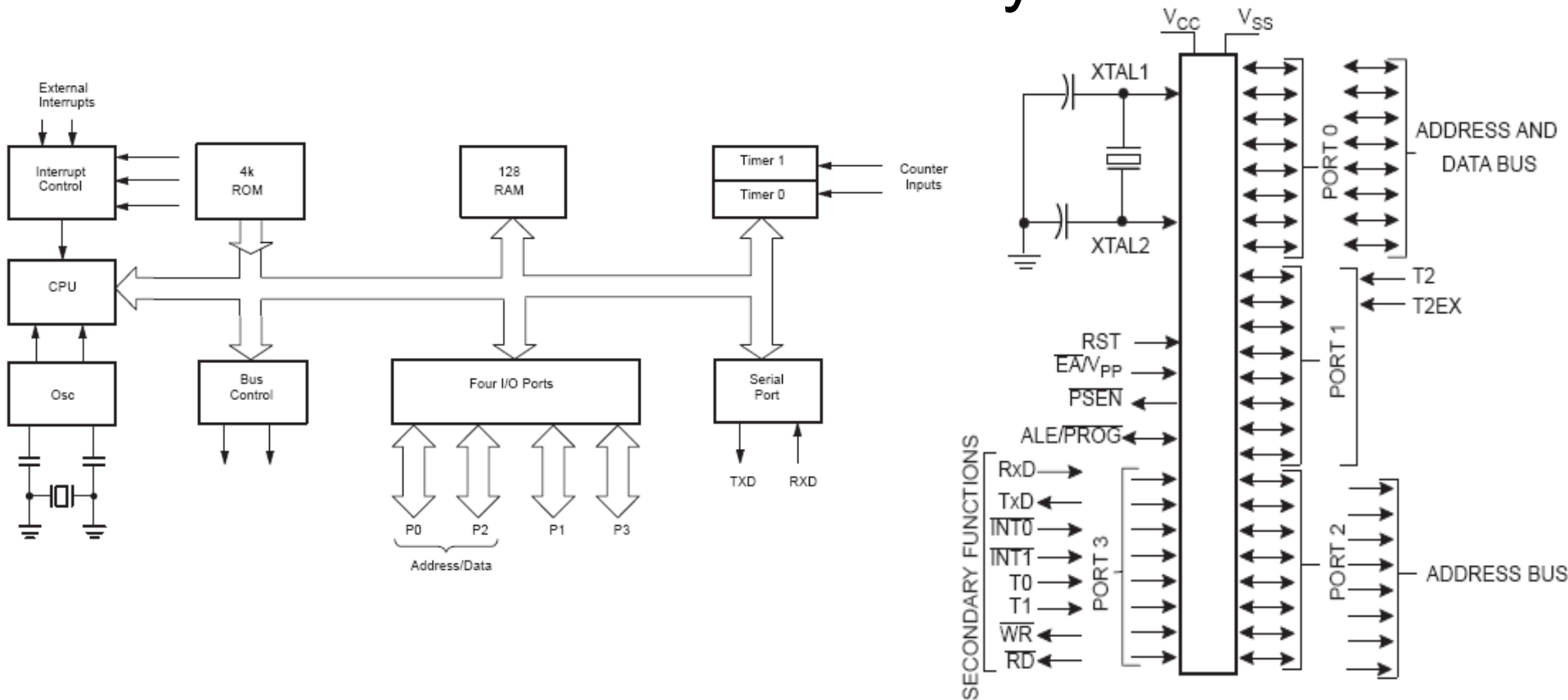
- Converters
 - Sampling
 - Nyquist criterion must be met: the sampling rate must be at least twice the highest baseband frequency:
 - Signal digitization
 - Analog-to-Digital Converters
 - Digital-to-Analog Converters

Need of conversion

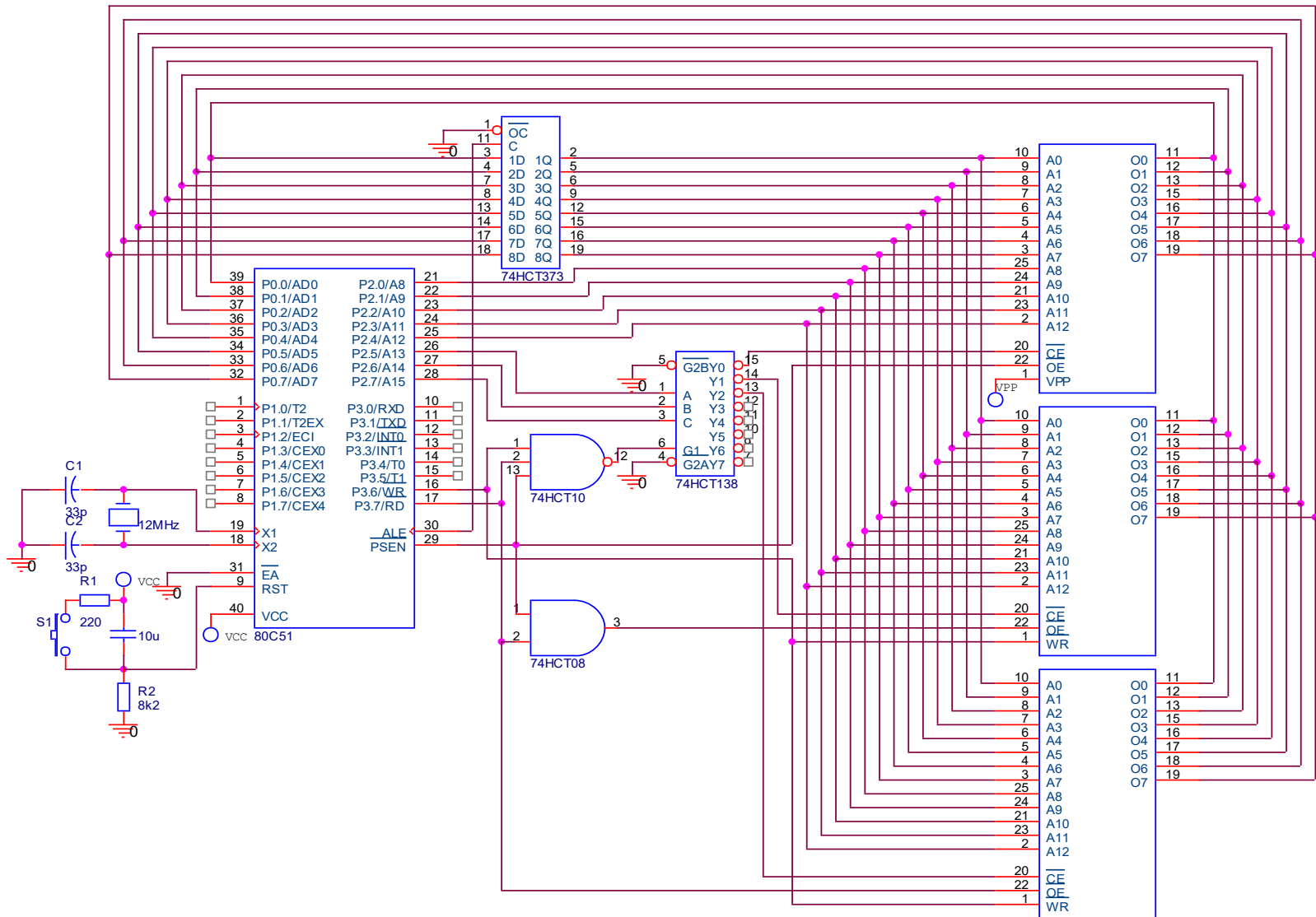


L11 Content

- 80C51 microcontroller family



L12 Content



Exam

- The **ADC exam** will take place in the laboratories of the **Observator** building, **second floor** on **31.01.2025**.
- Groups 30421, 30422, 30423/1: at 17:00
- Groups 30423/2, 30424, 30425: at 18:00
- **Details: on Moodle**

Exam

- **Moodle test:**
- 15 multiple choice items (with a single answer or with several correct answers; the number of correct answers is specified in the item)
- items are formulated from theoretical contents, respectively problems
- Test duration:
 - 50 minutes

ATTENTION: within an item/question, with correct answers you gain points, and with wrong ones you lose points. Don't choose the answers at random, because wrong answers cancel out right answers!

Exam

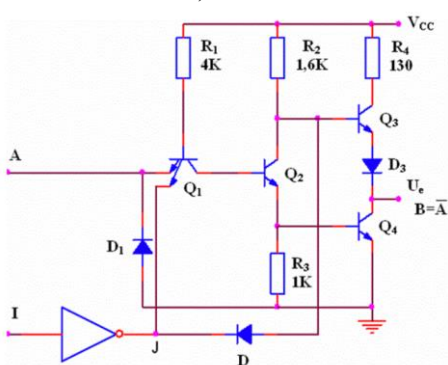
- **IMPORTANT:**

- the items are randomly chosen by the Moodle platform for each student (from a wide set of items)
- time management is up to each student (there is no time limitation per item, only a time limitation for the entire test); spend more time on the item containing the microcontroller problem (20 minutes)
- when the time expires, the test will end, and the answers will be sent automatically (even if you have not answered all the items)
- for each item a maximum of 0.6 points can be obtained, except for the item containing the microcontroller problem, for which a maximum of 1.6 points can be obtained.

The exam can only be taken from the labs (outside IP addresses will be monitored and blocked).

Exam questions example

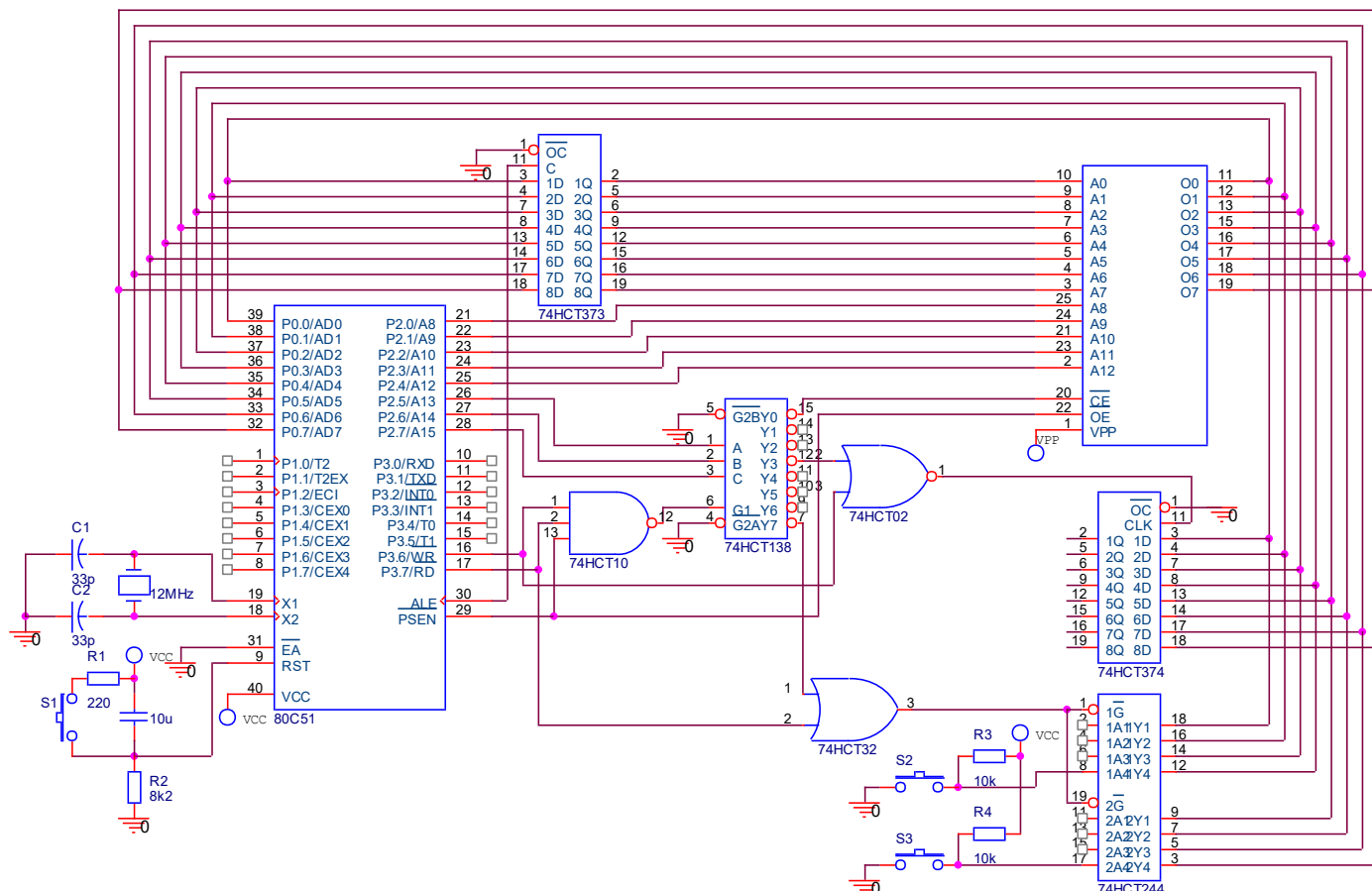
- Bifati varianta/varianțele corecte referitoare la filtrele trece sus și trece jos: (alegeți 1 răspuns)
Check the correct option (s) for the high-pass and low-pass filters: (choose 1 answer)
 - Filtrul trece sus atenuează frecvențele joase și alterează mai puțin frecvențele înalte; The high-pass filter attenuates the low frequencies and less alters the high frequencies.**
 - Reactanța capacitivă a condensatorului scade odată cu creșterea perioadei; The capacitive reactance of the capacitor decreases as the period increases.
 - La ambele filtre, distorsiunile introduse de circuit asupra semnalului de intrare pot fi reduse dacă constanta de timp RC este mai mare decât durata impulsului; In both filters, the distortions introduced by the circuit on the input signal can be reduced if the time constant RC is greater than the pulse duration.
 - Raportul de divizare depinde de frecvența doar în cazul filtrului trece sus; The dividing ratio depends on the frequency only for the high-pass filter.
- Caracteristicile amplificatoarelor operaționale sunt: (alegeți 2 răspunsuri)
The characteristics of operational amplifiers are: (choose 2 answers)
 - Impedanța de intrare foarte mare (ideal ∞); Very high input impedance (ideal ∞)**
 - Curenți de intrare foarte mici (ideal 0); Very small input currents (ideal 0)**
 - Tensiunea între intrări foarte mare (ideal ∞); Very high difference between the values of input voltages (ideal ∞)
 - Amplificarea sau câștigul în tensiune (AV) foarte mic (ideal 0); Very small amplification or voltage gain (AV) (ideal 0)
- În imaginea de mai jos, considerând o valoare a intrării I echivalentă cu "1" logic, alegeți afirmațiile care sunt corecte: (alegeți 2 răspunsuri)
In the image below, considering a value of input I equivalent to logic "1", choose the statements that are correct: (choose 2 answers)



- Q3 va fi în regiunea de blocare**
Q3 will be in the cut-off region
- Circuitul va prezenta la ieșire o stare de impedanță ridicată (starea HZ)**
The circuit will present at output a state of high impedance (HZ state)
- Q3 va fi în regiunea activă inversă
Q3 will be in the inverse active region
- Circuitul va prezenta la ieșire starea logică „1”
The circuit will present at output a state of logic “1”
- Q3 va fi în regiunea activă normală
Q3 will be in the forward active region
- Dioda D va fi blocată
Diode D will be blocked

- Ce componente înglobează un microcontroller? (Selectați două răspunsuri)
What components does a microcontroller include? (Select two answers)
 - Memorie; Memory**
 - Numaratoare; Counters**
 - Tastatură; Keyboard
 - Sursa de alimentare; Power supply
 - Ecran; Display

5. Se considera sistemul cu microcontroller 80C51 din figura. Care afirmatii sunt corecte? (alegeti X raspunsuri)
 Consider the 80C51 microcontroller system in the figure. Which statements are correct? (choose X answers)



- Circuitul generator de tact este format atat din componente externe microcontroller-ului cat si din componente integrate in acesta
 The clock generator circuit consists of both external components of the microcontroller and components integrated in it
- Sistemul are un port de iesire care poate fi scris prin intermediul unei variabile localizata la adresa 61B7H
 The system has an output port that can be written through a variable located at address 61B7H
- Pentru activarea portului de intrare se foloseste selectia partiala
 The partial selection is used to activate the input port
- Sistemul are un port de intrare care poate fi citit prin intermediul unei variabile localizata la adresa 84A1H
 The system has an input port that can be read through a variable located at address 84A1H
- Microcontrollerul foloseste doar memoria interna de program; The microcontroller uses only the internal program memory
- Pentru activarea portului de intrare se foloseste selectia totala
 The total selection is used to activate the input port
- Sistemul are un port de iesire care poate fi scris prin intermediul unei variabile localizata la adresa A475H
 The system has an output port that can be written through a variable located at address A475H

64k*8	A15	A14	A13	Hex	DCD
8k*8	0	0	0	0000	Y0
8k*8	0	0	1	2000	Y1
8k*8	0	1	0	4000	Y2
8k*8	0	1	1	6000	Y3
8k*8	1	0	0	8000	Y4
8k*8	1	0	1	A000	Y5
8k*8	1	1	0	C000	Y6
8k*8	1	1	1	E000	Y7